



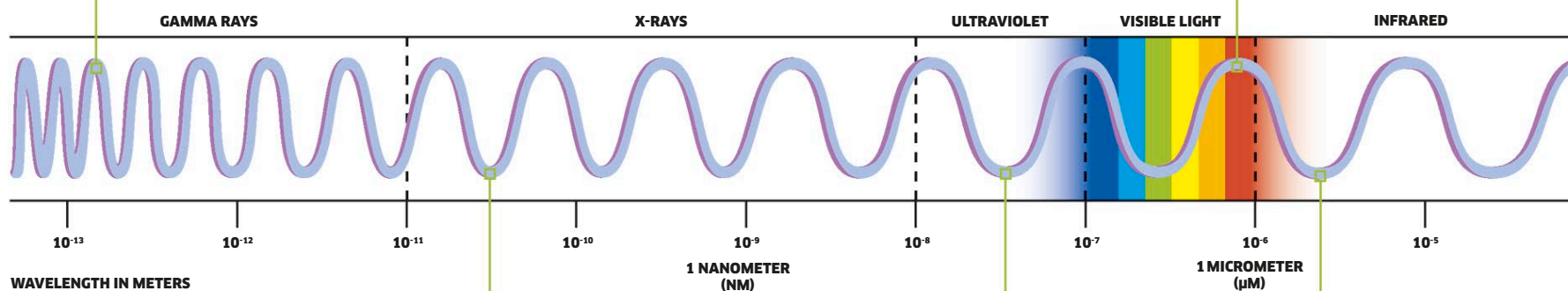
Gamma rays

The highest-energy waves, with wavelengths the size of an atomic nucleus, gamma rays are emitted by nuclear fission in weapons and reactors and by radioactive substances. Gamma radiation is very harmful to human health.



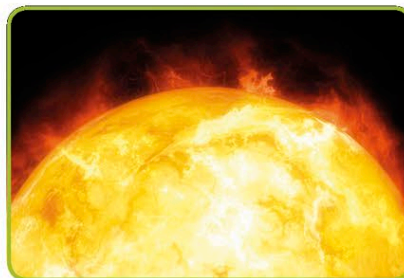
Visible light

This is the range of wavelengths that is visible to the human eye. Each drop in a raindrop is like a tiny prism that splits white light into the colors of the spectrum.



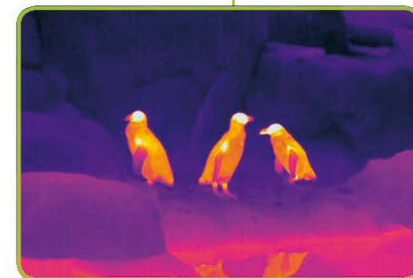
X-rays

With the ability to travel through soft materials but not hard, dense ones, X-rays are used to look inside the body and for security bag checks.



Ultraviolet (UV)

Found in sunlight, UV radiation can cause sunburn and eye damage. The shortest, most harmful wavelengths are blocked by the ozone layer.



Infrared

Known as heat radiation, infrared is invisible, but special cameras are able to detect it and "see" the temperature of objects such as these penguins.

Electromagnetic radiation

Light is one of several types of wave energy called electromagnetic radiation, which also includes radio waves, X-rays, and gamma radiation.

Electromagnetic radiation reaches us from the sun, stars, and distant galaxies. The Earth's atmosphere blocks most types of radiation, but allows radio waves and light, which includes some wavelengths of infrared and ultraviolet, to pass through.

The electromagnetic spectrum beyond visible light was discovered between 1800, when British astronomer William Herschel first observed infrared, and 1900, when French physicist Paul Villard discovered gamma radiation.

